

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

16th March 2018

Subject SA1 – Health and Care Insurance

Time allowed: Three hours (14.45* - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. * You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
- 3. You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
- 4. The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
- 5. Attempt all questions, beginning your answer to each question on a separate sheet.*
- 6. Mark allocations are shown in brackets.*
- 7. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** A new digital health insurance company wants to start a business of dental insurance covering all the treatments related to teeth of the insured. Company has aggressive business plans and wants to issue the policy through mobile application within 2 minutes. Company has empanelled dental clinics located in various parts of the country for treatment. The treatment cost is cashless and hassle free from policyholder's point of view.
- i)** Before launching this product what considerations will be taken into account by the company? (14)
 - ii)** Identify the risks around this product and how do you mitigate the risks. (10)

After doing the experience analysis, you have found that the overall claim experience across the portfolio of this business has not been favourable. Hence, it has been decided to increase the premiums across whole portfolio.

However, for age band 20-40 of male policyholders in rural locations, claim experience has been very favourable. Your marketing director was impressed with this statistics and he has proposed that, instead of revising the premium rates upwards across portfolio, premiums for each individual should be derived automatically using Artificial Intelligence system thereby policyholder with healthy tooth will be paying lower premiums compared to other policyholders.

- iii)** What considerations will you take before adopting the new technology? (8)

One toothpaste company has approached your company to market your dental insurance product and discount vouchers are offered with the purchase of toothpaste. Customer can buy insurance at discounted premium through using discount voucher.

- iv)** Discuss this proposal. (8)
- [40]**

- Q. 2)** A private health insurance company has only been selling indemnity health insurance for the last 8 years since its inception. The company's management is concerned as there has been a significant reduction in the new business volume over the last six quarters.

- i)** Discuss the probable reasons for reduction in new business for the insurer. (3)
- ii)** List the factors that influence the demand of private health care insurance (3)

The Government of the country has decided to implement a universal health care program whereby approximately half of the country population will be offered medical care.

- iii)** Discuss the probable impact on the demand of private health insurance once the program is fully implemented and resultant cost of medical health care as well as insurance premiums. (12)

The claims experience has also worsened over the period in question. An analysis of claims suggested that there are certain policy conditions that are vague and are being exploited by policyholders for making claims. The claims team has proposed that if these policy conditions are changed at the time of renewal then it may improve the overall claims experience.

- iv)** Discuss the factors to consider before changing the policy conditions for existing policies. (10)

The business intelligence team has come up with a proposal that there is an opportunity to increase new business by offering easier on-boarding in terms of relaxed underwriting requirements to individuals over the age of 50 years without changing the premiums. Insurance companies usually have very stringent underwriting requirements for people aged over 50. It is believed that stringent underwriting also dissuades healthier lives from opting for insurance. Hence by relaxing the underwriting requirements, the insurer will be able to attract a bigger pool of lives over the age of 50 years. This will not only increase the overall revenue but also help in managing the experience at the portfolio level.

- v)** Discuss the main factors that influence the approach to underwriting health insurance products. (8)
- vi)** List the factors that influence the propensity to claim by an individual. (5)
- vii)** List six actions that can be taken to deal with increased risk as a result of medical assessment. (3)
- viii)** Discuss the proposal in detail, its implication on the insurer's overall portfolio, possible risk mitigation measures and your recommendation (10)

The marketing team also suggested that if premiums for individuals in the age band between age 50 and age 65 are reduced compared to the industry, then it will also help in attracting healthier lives in these age bands. Relaxed underwriting along with lower premium would increase the sales.

The actuary estimates that it will require the company to cross-subsidize the younger lives for older lives by increasing the premium for younger age band and reducing for older age bands.

- ix)** Discuss the proposal and make your recommendation. (6)
- [60]**
